

Requirements Table — Student Startup Crowdfunding Platform

PES University – Lab 1: Requirements Engineering & UML Use-Case Modelling | Problem Statement #40 (Retail, E-Commerce & Finance)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall collect backer pledges and disburse funds to the student venture in tranches only upon verification of predetermined project milestones.	High	Pass: Milestone evidence is reviewed and the corresponding tranche is released. Fail: Campaign funds are disbursed without milestone sign-off.	Protects backers from misuse of funds and ties disbursement directly to verified progress. (Given)
FR-002	Functional	The system shall allow a student founder to create a campaign with a funding goal, campaign duration, a milestone schedule, and at least one reward tier.	High	Pass: A published campaign displays a goal amount, a start/end date, at least one milestone, and at least one reward tier. Fail: A campaign is published missing any of these fields.	A complete campaign profile is a prerequisite for accepting pledges and for later milestone verification.
FR-003	Functional	The system shall allow a campaign backer to select a reward tier at the time of pledging, restricted to tiers whose minimum threshold is met by the pledge amount.	Medium	Pass: Only reward tiers with a threshold less than or equal to the entered pledge amount are selectable. Fail: A backer can select a tier whose threshold exceeds their pledge amount.	Reward tiers are the primary backer incentive and must map correctly to contribution size to remain fair and enforceable.
FR-004	Functional	The system shall allow a student founder to submit evidence (documents, images, or reports) against a specific milestone for faculty committee review.	High	Pass: Submitted evidence is attached to the correct milestone and automatically generates a pending review request. Fail: Evidence is submitted but no reviewable record is created.	Committee review depends on structured, milestone-linked evidence rather than ad-hoc communication.
FR-005	Functional	The system shall allow a faculty committee member to approve or reject submitted milestone evidence, requiring a written comment whenever the decision is a rejection.	High	Pass: A rejection cannot be saved without a comment, and an approval automatically triggers the fund tranche release use case. Fail: A milestone decision is recorded with no reviewer comment on rejection, or approval does not trigger release.	Ensures founder accountability and gives founders actionable feedback when a milestone is not accepted.
NFR-001	Nonfunctional	The payment processing gateway must comply with PCI-DSS guidelines, using tokenized payment authorizations for all pledge transactions.	High	Pass: Benchmarking tests confirm the target latency and security standards are met under simulated peak load. Fail: Any raw card data is stored or transmitted unencrypted.	Handling real monetary pledges requires industry-standard payment security to protect backers and preserve platform trust. (Given)
NFR-002	Nonfunctional	The system shall maintain a tamper-evident audit log of every milestone verification decision and fund disbursement, retained for a minimum of 7 years.	High	Pass: Audit entries are hash-chained such that any modification or deletion is detectable, and a queried record returns within 2 seconds. Fail: An audit entry can be altered or removed without producing a detectable integrity failure.	A crowdfunding platform handling real money needs a verifiable trail for dispute resolution and financial compliance.

Student Startup Crowdfunding Platform — Use-Case Diagram



Use-Case Flow Specification

Student Startup Crowdfunding Platform — PES University Lab 1 (Problem Statement #40)

Use Case	UC-07: Review & Release Funds
Primary Actor	Faculty Committee
Secondary Actors	Student Founder (submits evidence), Payment Gateway (executes tranche transfer)
Includes	UC-08: Verify Milestone Evidence («include»)

Preconditions

- The campaign is active and has at least one defined, unreleased milestone.
- The student founder has submitted evidence for the milestone under review (UC-05).
- The reviewing faculty committee member is authenticated and assigned to the campaign's review queue.

Postconditions

- Success: The milestone is marked “Approved,” the corresponding fund tranche is released to the student founder's account, and both parties are notified.
- Failure: The milestone is marked “Rejected” with a reviewer comment, no funds move, and the founder is notified to resubmit evidence.

Main Success Scenario

1. Faculty committee member opens the review queue and selects a pending milestone submission.
2. System displays the milestone details and the founder-submitted evidence (UC-08: Verify Milestone Evidence is included here).
3. Committee member examines the evidence against the milestone's predetermined acceptance criteria.
4. Committee member selects “Approve” and confirms the decision.
5. System validates that the committee member has authority over this campaign's review.
6. System marks the milestone as Approved and calculates the tranche amount due.
7. System instructs the Payment Gateway to transfer the tranche to the student founder's linked account.
8. System logs the decision and disbursement in the tamper-evident audit trail (NFR-002).
9. System notifies the student founder and campaign backers that the milestone was approved and funds released.
10. Use case ends successfully.

Alternate Flow — 4a. Evidence Rejected

- 4a1. Committee member selects “Reject” instead of “Approve.”
- 4a2. System requires a mandatory written comment explaining the rejection before the decision can be submitted.
- 4a3. System marks the milestone as Rejected, logs the decision in the audit trail, and does not contact the Payment Gateway.
- 4a4. System notifies the student founder of the rejection and the reviewer's comment, allowing evidence to be resubmitted (returns to UC-05).
- 4a5. Use case ends without fund release.